

Python and SQL Foundations Code Portfolio

Business analytics programming fundamentals and relational query practice

Summary

This code portfolio documents introductory programming and relational querying work. It emphasizes technical fundamentals that support later analytics projects: defining inputs, manipulating data, using control structures, writing functions, querying relational data, and communicating code clearly through comments and documentation.

Area	Representative work	Portfolio signal
Python basics	User input, calculations, formatted output	Translates problem statements into executable logic.
Control flow	if/elif/else, for loops, while loops	Writes programs that branch and iterate over data.
Functions/lists	Reusable functions, min/max, mean and standard deviation	Modularizes code and performs simple analysis.
SQL foundations	SELECT, WHERE, GROUP BY, JOIN, summary queries	Queries relational data and creates business summaries.

Selected Python Code Pattern

```
def calcCost(count, price, discount=0.10):
    if count >= 5:
        total = count * price * (1 - discount)
    else:
        total = count * price
    return total

def minmax(values):
    low = values[0]
    high = values[0]
    for value in values:
        if value < low:
            low = value
        if value > high:
            high = value
    return low, high
```

This example demonstrates functions, parameters, conditional logic, list iteration, and returning calculated values.

Selected SQL Code Pattern

```
SELECT
    c.Country,
    COUNT(i.InvoiceId) AS invoice_count,
    ROUND(SUM(i.Total), 2) AS total_revenue
FROM customers AS c
JOIN invoices AS i
    ON c.CustomerId = i.CustomerId
GROUP BY c.Country
ORDER BY total_revenue DESC;
```

This example demonstrates joins, grouping, aggregation, aliases, and sorting to produce a business-style country revenue summary.

Skills Demonstrated

- Programming logic and problem decomposition
- Python data types, lists, loops, functions, and basic statistics
- SQL table querying, aggregation, and joins
- Code comments and learning documentation
- Organizing code for public portfolio review

Limitations and Scope

This is a foundations project. It is not presented as production software, a deployed application, or an advanced machine-learning workflow. Its purpose is to document baseline technical fluency that supports the more advanced analytics, forecasting, econometrics, and consulting projects elsewhere in the portfolio.

Workflow Visual

Python and SQL Foundations Workflow

Programming portfolio artifact

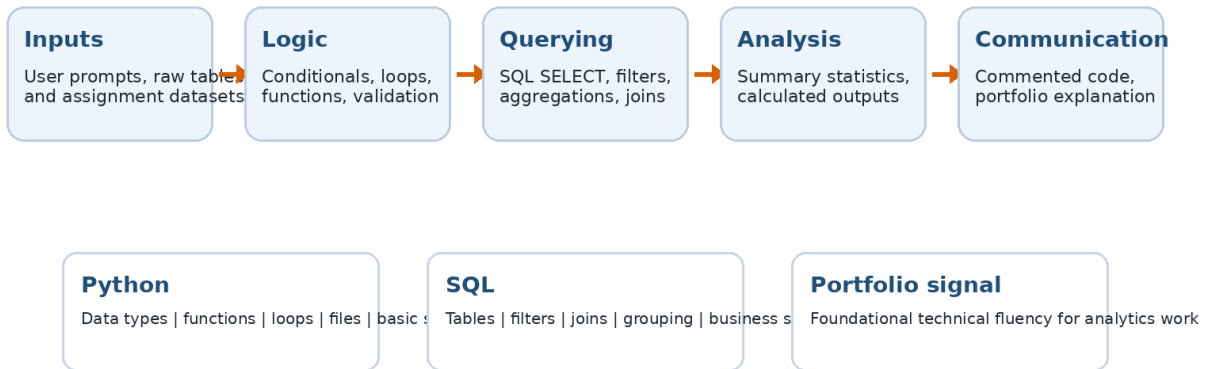


Figure: Workflow connecting programming fundamentals, relational querying, and analytics communication.

Figure: Programming and querying workflow used to frame the foundations project.